

Full Stack Development with MERN

Project Documentation Date:

17 February 2026

1. Introduction

Project Title: HouseHunt – Finding Your Perfect Rental Home

Team ID : LTVIP2026TMIDS81826 Team

Members:

Team Leader : Katta Aakash

Team member : Koruprolu Satya Divya

Team member : Sai Kammila

Team member : Sai Pradeep Bolla

2. Project Overview

Purpose:

HouseHunt is designed to provide a secure, scalable, and user-friendly rental property management platform. The system ensures verified owner onboarding, transparent booking workflows, and structured administrative governance. The platform eliminates traditional inefficiencies in rental systems by digitizing the end-to-end lifecycle from property listing to booking confirmation.

Features:

The application includes secure user registration and login, role-based authentication for Admin, Owner, and Renter, property CRUD operations for owners, structured booking request and approval workflow, and admin-controlled owner verification system. Each module is implemented following REST architectural principles.

3. Architecture

Frontend:

The frontend is developed using React.js with functional components and hooks. React

Router is used for client-side routing, and Axios is used for secure API communication. State management is handled using React's built-in hooks, ensuring modular and maintainable UI design.

Backend:

The backend is built using Node.js and Express.js. RESTful APIs manage authentication, property management, booking lifecycle, and admin approval mechanisms. Middleware is used for JWT verification and role-based authorization.

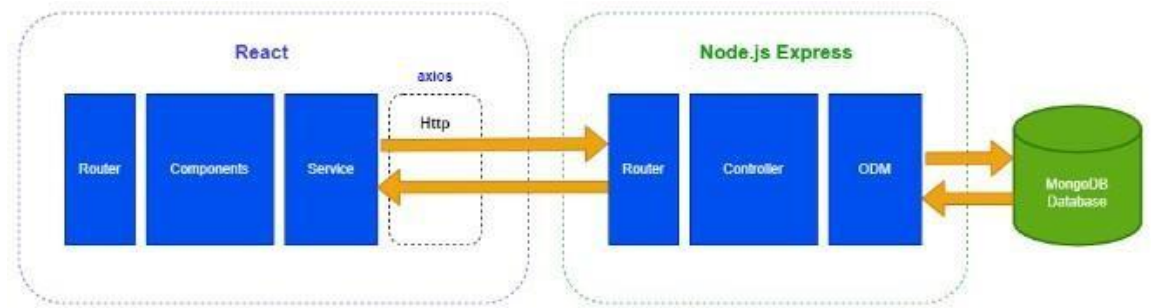
Database:

MongoDB serves as the NoSQL database. Mongoose ORM is used for schema modeling and validation. The system uses three primary collections: Users, Properties, and Bookings.

Database Schema Overview:

Collection	Key Fields	Description
Users	userId, email, password, role	Stores renter, owner, and admin accounts
Properties	propertyId, ownerId, rent, location	Stores property listings
Bookings	bookingId, userId, propertyId, status	Stores booking requests and status

System Architecture Diagram:



4. Setup Instructions

Prerequisites:

- Node.js
- MongoDB - npm

Installation:

Step 1: Clone the repository git clone

<https://github.com/pradeep4129d/househunt>

Step 2: Backend Setup

Create an env file in the backend root folder and add MongoDB connection URL

cd server npm install npm start

Step 3: Frontend Setup

cd client npm install

npm start

5. Folder Structure

frontend Directory: Contains reusable components, pages, services for API calls, and main App.js entry point. Follows modular component-based architecture.

Ex:

frontend/src/components

frontend/src/context frontend/src/styles

backend Directory: Organized into models (database schemas), routes (API endpoints), controllers (business logic), middleware (authentication & authorization), and server.js (application entry point).

Ex:

Backend/models

Backend/routes

Backend/server.js

6. Running the Application

To run the backend, navigate to the server directory and execute npm start. To run the frontend, navigate to the client directory and execute npm start. The frontend communicates with backend APIs running on configured port.

7. API Documentation

The backend exposes REST APIs for user management, property management, and booking lifecycle management. Each endpoint follows standard HTTP methods such as GET, POST, PUT, and DELETE.

API Endpoint Overview:

Method	Endpoint	Description
POST	/api/register	Register new user
POST	/api/login	Authenticate user
GET	/api/properties	Fetch properties
POST	/api/property	Add new property
POST	/api/booking	Create booking request

8. Authentication

Authentication is implemented using JSON Web Tokens (JWT). Upon successful login, a signed token is generated and stored on the client side. Middleware validates tokens for protected routes. Passwords are hashed using bcrypt before storage to ensure data security.

9. User Interface

The user interface is built using React.js and Bootstrap to ensure responsive design. Separate dashboards are provided for Renters, Owners, and Admin. The UI emphasizes usability, clarity, and accessibility.

- Registration Page
- Login Page
- Renter Dashboard - Owner Dashboard
- Admin Dashboard
- Property Listing Page
- Booking Management Page

10. Testing

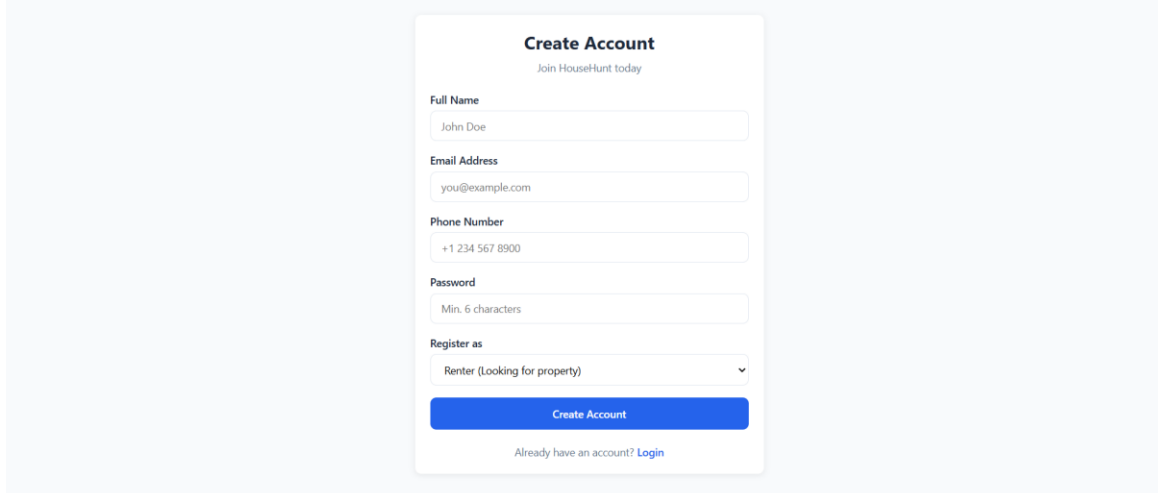
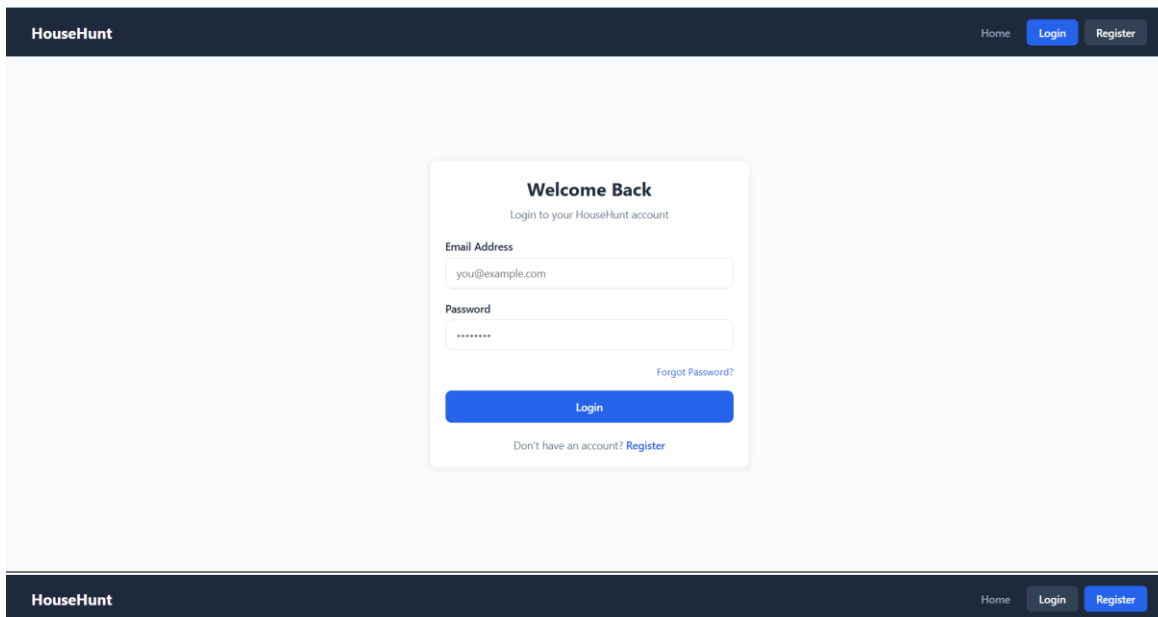
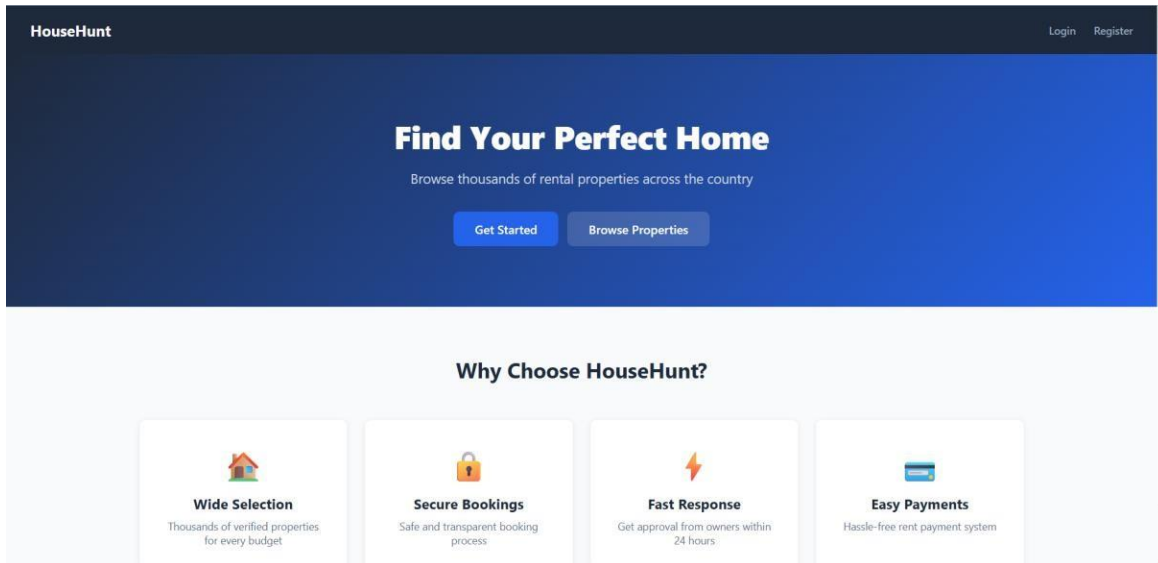
Comprehensive testing was conducted including API testing using Postman, functional testing for booking workflow, and role-based access validation. Edge cases and invalid inputs were tested to ensure robustness.

11. Screenshots or Demo

GitHub Repository: <https://github.com/pradeep4129d/househunt>

Demo

Link: <https://drive.google.com/file/d/1HKCZYdcZ39zthmswCL69hApZESza2yBY/view?usp=sharing>



Admin Side:

HouseHunt Admin

Dashboard

All Users

All Properties

All Bookings

Dashboard

2
Total Users

1
Owners

1
Renters

4
Total Properties

2
Available

2
Total Bookings

0
Pending

2
Approved

Quick Actions

Manage Users

Manage Properties

View All Bookings

System Info

Platform: HouseHunt v1.0

Role: Administrator

Email: admin000@gmail.com

HouseHunt Admin

Dashboard

All Users

All Properties

All Bookings

All Users

2 users

NAME	EMAIL	PHONE	ROLE	JOINED	ACTION
Kammila Sai	saikammila143@gmail.com	8179077852	Owner	2/18/2026	Delete
Kammila Sai	saikammila123@gmail.com	8179077851	Renter	2/18/2026	Delete

4 total

title	owner	city / state	type	rent/mo	status	action
Budget Studio Room	Kammila Sai saikammila143@gmail.com	Pune, Maharashtra	Studio	₹10,000	Rented	<button>Delete</button>
Luxury Villa with Private Garden	Kammila Sai saikammila143@gmail.com	Bangalore, Karnataka	Villa	₹75,000	Rented	<button>Delete</button>
Spacious 3BHK Family Home	Kammila Sai saikammila143@gmail.com	Hyderabad, Telangana	Apartment	₹15,000	Available	<button>Delete</button>
Modern 1BHK Near Metro	Kammila Sai saikammila143@gmail.com	Mumbai, Maharashtra	Apartment	₹17,999	Available	<button>Delete</button>

All Bookings

PROPERTY	RENTER	OWNER	START DATE	END DATE	AMOUNT	STATUS
Budget Studio Room Pune	Kammila Sai saikammila123@gmail.com	Kammila Sai saikammila143@gmail.com	2/18/2026	2/19/2026	₹10,000	approved
Luxury Villa with Private Garden Bangalore	Kammila Sai saikammila123@gmail.com	Kammila Sai saikammila143@gmail.com	2/18/2026	2/21/2026	₹75,000	approved

Rent Owner Side:

HouseRent Owner

Dashboard

Add Property

My Properties

Bookings

Add New Property

City

Mumbai

State

Maharashtra

Full Address

Street, Area, Landmark

Monthly Rent (₹)

15000

Property Type

Apartment

Bedrooms

1

Bathrooms

1

Area (sq ft)

800

Amenities (comma separated)

WiFi, Parking, AC, Gym

Property Images

Choose Files | No file chosen

Add Property

Cancel

HouseHunt Owner

Dashboard

Add Property

My Properties

Bookings

My Properties

+ Add Property



Budget Studio Room

Rented

Room 12, IT Park Road, Hinjewadi, Pune

₹10,000/mo

1 Bed • 1 Bath • Studio

Delete Property



Luxury Villa with Private Garden

Rented

Villa 12, Prestige Layout, Whitefield, Bangalore

₹75,000/mo

4 Bed • 3 Bath • Villa

Delete Property



Spacious 3BHK Family Home

Available

Plot 45, Banjara Hills Road No. 12, Hyderabad

₹15,000/mo

3 Bed • 2 Bath • Apartment

Delete Property



Modern 1BHK Near Metro

Available

HouseHunt Owner

Dashboard

Add Property

My Properties

Bookings

Booking Requests

PROPERTY	RENTER	DATES	AMOUNT	STATUS	ACTIONS
Budget Studio Room Pune	Kammila Sai 8179077851	2/18/2026 – 2/19/2026	₹10,000	approved	
Luxury Villa with Private Garden Bangalore	Kammila Sai 8179077851	2/18/2026 – 2/21/2026	₹75,000	approved	

HouseHunt Owner

Dashboard

Add Property

My Properties

Bookings

Owner Dashboard

Kammila SaiLogout

4
My Properties

2
Total Bookings

0
Pending Approval

Quick Actions

+ Add New Property

View My Properties

Manage Bookings

Account Info

Name: Kammila Sai

Email: saikammila143@gmail.com

Role: Property Owner

User side:

HouseRent

[Browse Properties](#) [Kamila Sai](#) [Logout](#)

My Dashboard

[Browse Properties](#)

2

Total Bookings

0

Pending

2

Approved

My Bookings

PROPERTY	DATES	AMOUNT	STATUS	ACTION
Budget Studio Room Pune • ₹10,000/mo	2/18/2026 – 2/19/2026	₹10,000	approved	
Luxury Villa with Private Garden Bangalore • ₹75,000/mo	2/18/2026 – 2/21/2026	₹75,000	approved	

HouseRent

[Dashboard](#) [Logout](#)

Browse Properties

2 properties found

City

Type

Min Rent

Max Rent

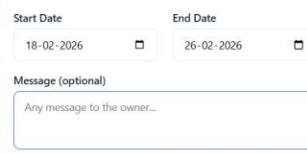
Search

Search city...

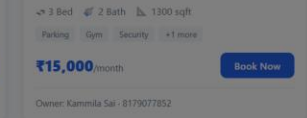
All Types



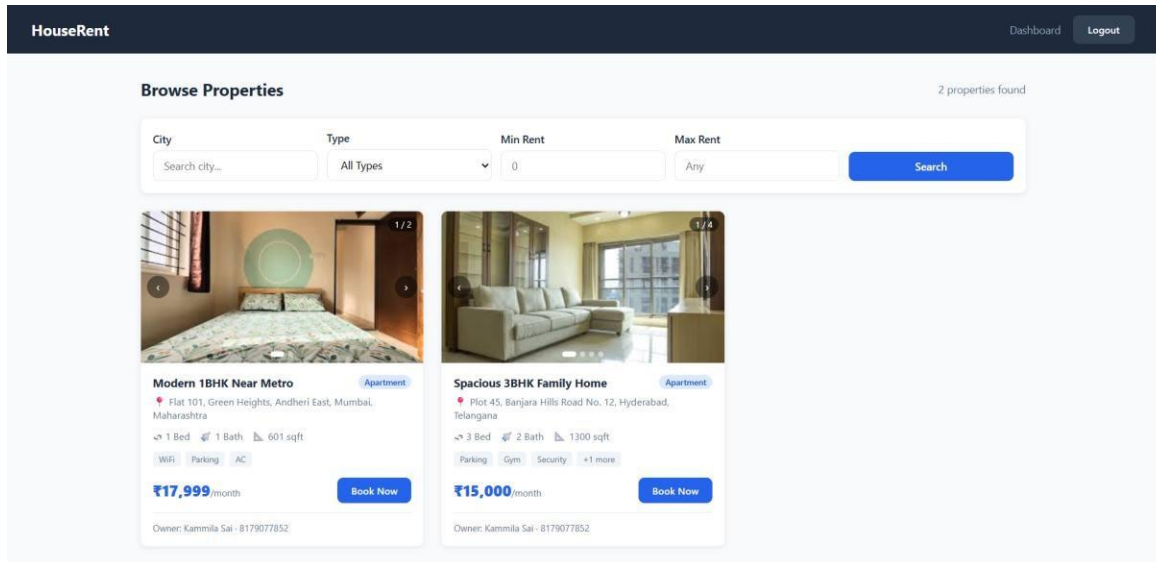
Modern 1BHK Near Metro [Apartment](#)
Flat 101, Green Heights, Andheri East, Mumbai, Maharashtra
1 Bed 1 Bath 601 sqft
WiFi Parking AC
₹17,999/month [Book Now](#)
Owner: Kamila Sai - 8179077852



Book Property
Spacious 3BHK Family Home
Start Date: 18-02-2026 End Date: 26-02-2026
Message (optional)
Any message to the owner...
[Send Request](#) [Cancel](#)



3 Bed 2 Bath 1300 sqft
Parking Gym Security +1 more
₹15,000/month [Book Now](#)
Owner: Kamila Sai - 8179077852



12. Known Issues

The system currently does not integrate online payment gateways or real-time chat modules. Email verification and advanced notification systems are not implemented.

13. Future Enhancements

Future enhancements include payment gateway integration, AI-based recommendation engine, real-time messaging using WebSockets, containerized deployment using Docker, and cloud deployment on AWS or similar infrastructure.